

"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 2

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Hello, it's July 1st, 2023. We're in Physics as Information Processing in the second discussion.

So thanks all for joining. We'll see who else comes in. If you're watching live, feel free to write questions in the live chat and I'll relay them onwards. So Andre, thanks. And it would be awesome to begin with a little bit of some overview and some points that you wanted to get to. Yeah, thank you, Daniel. So good morning, everyone. So if you would start by bringing up a bit cold with Chris's spoke of last time. I have a few slides that I singled out. So now we're getting to reformulation of quantum theory in terms of reformulation of opposed to dynamical theory. And as we've talked in previous sessions, we need to think a little bit philosophically what the ontology of measurement entails.

So Chris, last lecture started with these three basic ingredients.

So an action, that's number one, a thing to ask about. This is the qubit, the physical degrees of freedom that we need to perturb in some way to ask a question. And then finally, a semantics, a language that makes sense, a context that makes sense of that information that we gather. So I should say, number one, I came up in the first session when, I can't exactly remember, but at some point we talked about what is information. And we came up with, or we brought up a definition that goes something like this. It's a nice adage. It's differences that make a difference. This is, Chris told me, I can't remember who first coined this adage, but I think it's quite old, at least early, early 70s. Anyway, so number one, basically, let's curve that part, right?

If I want to gather some information, I need to perturb a system somehow. So I need to act. And this is literally in units of actions, right? So units of energy times time. And he provided the example of rhodopsin, right? Which is apparently near optimal from a quantum perspective. This is mechanically what goes on in your eyes as light comes in. And this is literally the difference that makes a difference in your brain. The second part were the physical degrees of freedom. So recall that we had a boundary, okay? So this theory of quantum mechanics as measurement necessarily breaks down the system into a bipartite system with a boundary. And you are in a boundary, which are a lot smaller than than the degrees of freedom of A and B, at least in terms of the dimension of the Hilbert space. And the other medium change, right? Sorry, can you repeat, Andre?

I repeat what? Sorry? Just last 30 seconds.

Yeah. So the second ingredient is the physical medium in which this information exchange occurs. So the physical degree of freedom, the qubits are arranged and these are physically perturbed in order to mediate that difference that makes a difference. So we can think of this duly as either preparing those qubits or decoding those qubits. And then finally, there was the notion of the reference frame.

Okay. So this theory necessitates a free choice of the agent of a reference frame. And physically, what this amounts to is a free choice of a measurement axis for those qubits. So we can think of measuring Z-spin in those qubits on the boundary. So, you know, that requires a free choice of that axis.

And this, you know, amounts to your reference frame, the context in which those measurements are performed.

On one extreme case, we have that if those references are perfectly aligned, then the exchange of information is noiseless. There was somewhere over here, I think.

I can't quite recall. But generically, these two reference frames of Alice and Bob A and B are not exactly the same. They get to choose the reference frames. I think it's all the way on the bottom, actually. Right? So they can be aligned. If they're aligned, they are exchanging noiseless information. Right? There is no randomness. If they are not superposition, and that's when you're going to get, you know, probabilities in the outcomes, trivial probabilities. So Marcia, good?

Yes.

Okay. So I have a few more comments of the main things. Okay. It'll take me a few minutes, and then we can chat back and forth. Perhaps this slide is the most interesting, important. I remember this is where I asked the question myself because I was, I still, you know, it's not easy to wrap your head around what's going on here.

So we said that measurement and preparation are completely dual pictures.

So this is going to be expanded on in the next lecture. But a direct corollary of that is that if you prepare, Chris said it in similar terms, if you prepare something in a certain state, and then you measure it right away, you should expect that it's still in that state. Right? So, intuitively speaking, this induces a cycle, right? And where you have preparation and measurement, and you're back to square one, so to speak. And in generating that cycle, obviously, there has to be some notion of time. Right? So that's where we're going to have to make sense of the distinction between internal and external time. If we take this measurement operator, and here perform some operations, we can write it as a unitary propagator from quantum mechanics, which again preserves information. It just leaves the state as it is. But with this little trick of the weak rotation, you can write it as e to the measurement up to a constant. And that, you know, precisely encodes, at least morally speaking, what we just said, right, that this preparation measurement thing is a cycle. And more importantly, it gets rid of this t external time. And presumably, in here, it's going to be replaced by some notion of internal time, we're going to have an internal time QRF, that is sort of quote, unquote, the true local time, as the observer proceeds.

I'm not entirely certain of the details. But now looking at it here, the beta, which presumably there's a, you know, beta and inverse temperature here, I think this is going to be intimately related to notions of thermal time. This has been explored in papers by Rovelli.

Call, you know, internal time QRFs and thermal time is the way in which the observer observes the passage of time. Those papers of Rovelli are linked in the Q&A, by the way, so I can also provide links later. I think there might have been a question. Should I stop?

Yeah, Dean, go for it.

I was curious about this, because what I heard Chris start this session with was that he was going to try to

help us understand when we are looking at something through a mechanical lens. So quantum mechanics. And to me, this slide was the way of addressing, as he said, his words were characterized information transfer in a quantum theoretic way, that the translation from H to M, took the mechanics aspect out of the change or timepiece. It was the purification, the most theoretical form of transfer of information, as opposed to holding up just the one ordered or sequenced or mechanical lens.

Did I hear you ask the question? And I kind of smiled because I thought to myself, no, I understand why Anders is asking this question. It's, can something be both gripper in this in the conceptual meaning sense ontologically, and gripped a physical object. And I think what I thought, maybe I was misinterpreting, I'd like your what your perspective on this is, that ability to go from H to M is what allows us to have both the mechanical and the theoretical, the transference in its purest sense. Is that being able to have both? Both can be true? Is that what it I think?

I think the answer is, yes, for sure. Yeah. Yeah. Yeah. I

I think much else to say here. It's, it is, it is true that you can think of the information transfer picture as an enlargement or an addition to the mechanical picture. And that I don't think that the mechanical picture has to be just, you know, thrown out completely. I think, I think it's interesting to keep it to see how it can be recovered from the information processing picture. This reminds me, actually, there is an Italian physicist, I think his name is Cribella. And Dean, if you want, I can, I can link you the papers later, who is very fond of the information theoretical approach to quantum mechanics and deducing quantum mechanics from first principles. And he has a paper. I think the title is physics without physics or something like that, where, you know, he, he treats, um, uh, this whole thing as a seller, big seller automaton. And by taking the appropriate scaling limits, he derives QFT. I'm interacting QFT. So, so he says, okay, let me start with some axioms, how information should be the information processing. I mean, I'm very, very loose here, then get a cellular automaton out of this. And then take scaling and recover QFT, non interacting QFT, at least. So, you know, start with a few axioms, a lot of information processing, recovery mechanics from that. Does that add to your question a little bit? Yeah, so, so all I'm, all I'm trying to do is, is tie this back into some of the live streams that we've done in the past. And I know I use different language. I, I am not a scientist in the, I've not studied quantum other than sort of in passing ways. But I do think that what this spoke to me about was something that I've always tried to include, which is that there's the figure it out piece, which to me is the conceptual and the meaning part that Chris talked about very clearly here. And then there's the outing of the figure, there's the sort of physical edge and boundary that if you don't have an awareness of both things happening

concurrently, the preparation and the measure are unitary, have sort of minimum of two state, then all bets are off in terms of being able to think of these, these propagators and operators and all of the ways that we're trying to explain the situation, they're off. So we have to have both.

We have to have that sense that there is a transfer, but there are, and there is the mechanics, we have to keep both juggling all the time. So that's where I, that's where this one particular slide was really quite both bias confirm confirming for me, but I think it also, as you say, is going to open up a lot

that July. Yeah, I agree. I think this is the most, you know, important or mysterious if you wish a slide. Yeah. And it turns the pyramid, the conventional approach, here's a pyramid into a top that can be spun essentially with a rotation and it, and it validates how information transfer probably gets short shrift in most of this quantum conversation, but it's, it's probably at least an equal to, to the mechanical part that what, what order are we delivering this information in? So I'm at that point, I'll stop. Thanks. Yeah.

Thanks. Alexi.

Hi, can you hear me?

Yep.

So, I mean, I, I'm an, you know, my mathematics teacher in high school used to say, this is beautiful, but I don't quite understand it. So I wanted to, maybe I have two questions, but I'll start with, with this simpler one. So when Chris talked about the reference frame and said common language, you know, I think in Shannon common language is the alphabet. There's a receiver and sender and the alphabet that they share. And then Chris said, this reference frame is a physical thing like neurons or hands or mouth. And then I got lost because, you know, if, uh, when I encode, I modify something in the, on the hard drive, then the physical components of the hard drive is the reference, uh, frame. But, you know, this is not the shared language, you know, so that's my first question. Maybe I'll stop here to, to really, I understand we're dealing in metaphors, but I want to understand that, you know, the reference frame up or down, if Bob and Alice understand it similarly, then is that really physical?

I think the answer is that

Chris would probably give a better answer.

Start from the beginning.

Go.

Is that both are true at the same time. I think Chris would give you a better answer, but the, the semantics or the, sorry, the shared language or reference frame has to have a physical embodiment. And, and that's a hardware, a piece of hardware. It's, it's a neuron or something. Um, in this case, it's the, the, your free choice of the spin axis and that, that's your reference frame. But at the same time, it's, it's a physical, it's a physical piece of hardware. So reference frames are both the piece of hardware, but also a piece of semantics.

So if I take it from the language, I don't quite understand to computer science. So if Alice and Bob communicate through a piece of memory, whether it's random access memory, hard drive, then Alice encodes something on that hard drive and Bob reads from the hard drive. So in that sense, they understand how to encode and decode, but they use the same physical piece. That's the reference frame. Would that be correct?

I might, I might just add one again, continuing with the computer science, let's just, and the Shannon alphabet. Let's just say that these are plain text character encodings. And so they share the same

set of character, a through Z encodings that, that is reflected somewhere physically.

And so if, if a message were, if, if, if a message were sent from another letter, it wouldn't render properly. It's like outside, they couldn't differentiate Omega from Zeta because they only have the A through Z software pack. And so there's an informational distinguishments that's facilitated by the information that's classically inscribed on the screen, the memory. And there is, is an embodied physical as well as a semantic aspect to that situation.

All right. I think I'm getting closer to understanding. So to move away from computer science, if Alice is a painter, she took a brush, dipped it in red paint and painted on the canvas red. And Bob has the same understanding of, of red as Alice. And then he looked at the canvas and read, read that it is both the physical, you know, a mark on the canvas and, and the symbol of red. Right. Yes. And if, if, if somebody had red, green color blindness, that physical apparatus was not able to, to informationally distinguish, then they, they couldn't make this wick rotation into the quantum cognitive space. They wouldn't be able to, to, um, detect that there was a red figure painted on a green background.

Yeah.

Yeah.

Okay.

The key component here is that the, the reference frame is inside of the edge. It's, you know, if you're a color blind, there is something physical inside of your head or your eyes or whatever that does not allow you to see the difference. So, so your semantics are affected ultimately by a physical piece of hardware.

Well, yeah, maybe I used a bad example because color actually is computed in the brain and color blindness is not necessarily physical. So, uh, okay. But I, I think I get it closer, but you know, if we, if I may ask another question, I get lost with a saying that preparation and measurement are, they have to happen both. And that when you prepare or when you read, you modify. So I am okay with the act of writing, the act of encoding. So when I encode something such as I, you know, mark red on the canvas, or I write to the hard drive, a, I modify the hard drive. That is true. But if I, if I am asking a question, right, I'm not modifying anything on the canvas. So in what way the act of asking the question or preparation is modifying the qubit, this is where I'm lost. You know, if I, does that make sense?

Yeah. I think this is exactly the question I asked, uh, to Chris in, in the stream. Um, so I can't quite remember his answer, but I think it was basically that they're completely dual pictures. Um, in the sense of, you know, they're, they're both two ways of looking at the same thing. Uh, it's kind of like, you know, evolving towards the future or, or, or, or the path you have time symmetry, you know, you can, you can do either. Um, yeah. So I, I myself, I'm slightly confused by the very, but I think the answer is that they are anything at the same, the picture of, or am I preparing or just completely dual? Okay. That's it. Well, do you remember that, Daniel?

Yeah. Okay. I'll give a different angle than the, um, the formal duality relationship. So let's think

about our measuring device is going to be a carpet and we're interested in whether somebody has walked on the carpet. So to prepare the carpet, we comb it very nicely in one direction. And I'm sure there's analogies there with the, um, preparation of qubits and a quantum computer, but we're going to comb the carpet as an act of preparation. And then something is going to happen. Maybe it's a split second later. And so we're very confident that it should be still nicely combed, or maybe it's going to be a whole working day later, but now we're going to take that probe, the tip, whatever it may be, and we're going to run it along the carpet. And we're going to measure the state of, of the direction of the fibers of the carpet. So if we didn't prepare the measurements, we wouldn't have a calibrated reference point to know how things had even changed. And so there's, there's a cycle of preparing the apparatus through something like a calibration, or at least like an emptying of memory to be able to then make an observation and to have that observation, have the kind of meaning that we would have expected, like purge the photo detector, make the measurement. Now you have to purge it again.

That's the preparation. What do you think Alexei? That is a beautiful metaphor. And I dig that, except I think I heard Chris say that when you prepare, you may cube it into up or down. And this is when I'm lost because when you comb the carpet, you don't do that. You don't put the footstep on it. In fact, there may not be any footstep, right? And this is, this is where I'm missing, you know, I think in quantum physics as well, or if you measure there's the cause and effect relationships are broken here for me. You know, if in my act of preparation, I'm opening my canvas to, to read the message, but I do not know what the message will be, I'm game. But if he's saying that when I prepare, I modify the carpet such that it becomes either up or down, actually not blank, but up or down, then I'm lost.

You have to comb the carpet up or down. There's no blank carpet. So you, you, um, you know, you have a slate, you have an etch a sketch. You could either make all the pixels white, and then you'd be able to measure a message in the dark, or you can make all the message dark or the slate dark. And then you could be able to measure something that was light, but you pick a direction of polarization in your preparation or calibration step so that deviations from that can be the measurements.

So you're saying carpet is binary. Carpet doesn't have multiple states. Carpet can only have two states. So if I prepare it, I, I effectively put it in one of those states. Well, let's say anything.

Yeah. Let's say we had the carpet at a 45 degree angle. Mathematically, we could describe that as a mixed state and it may not be the most effective for measuring the movement of people across the carpet, but yeah, you could, um, define it, um, or comb it that way. It doesn't necessarily, if it, if it were only a binary, it wouldn't be a qubit, then it would be a classical bit.

I actually, yeah. Andre, go for it. And then Dean.

Yeah. With, with, with, you know, I, I, you know, recall lecture a couple of slides more.

We briefly talked about it like 10 minutes ago by preparing, you are choosing an axis and whatever we're done in that axis. Now that doesn't mean that the environment gets to do its own thing. Right. And a superposition of, uh, outcomes, but at least you're measuring that axis. Right. So it's, it's going to be either up or down in that, in, you can add it to picture. And I have, uh, I just want to say a few

words on graphic screens and define my diagrams and maybe clarify the question a little more.

Dean. Dean.

On this really, really interesting. I don't know how metaphorically you could explain the carpet as a wave, but I think once you get away from the, it's up or down, that possibility enters the conversation. And the only reason I, I wonder whether or not metaphorically the wave aspect of that can be included in how you're describing the carpet is that Chris spent quite a bit of time on the all possible paths argument. And in particular, and I love this when Alice isn't looking.

Dean. Right. Well, in that, in that wave equation, how could Alice not be looking?

That's what gets real. I mean, Alice could not be looking if it's, if it's binary, if it's up or down, but if it's wave, it's on a, it's on a, it's a unitary propagator. I'm not sure how she takes her eyes off of this preparation measure thing that's going off all the time. So this is, this is actually an, what I consider to be a feature or an opportunity or an opening

in the sense that from a strictly information transfer piece, not a mechanical piece. Now, not a physical thing.

From an information transfer piece, maybe the carpet can be metaphorically understood as a wave.

That the eyes of Alice or whoever's doing the preparing and the measuring, Bob, Alice, I don't care, peaches. That there is actually a difference then between something that is exchanged and something that is informationally transferred. And again, I don't think that the four of us here sitting right now are going to resolve that. But I wonder if part of our, part of the difficulty in this is

finding the physical metaphor that explains the theoretical transfer in such a way that it doesn't, it, it, it both breaks the symmetry and holds it. Because again, we're back to gripper and grip. We're back to figuring it out and outing the physical figures in the state that we're in. And we have to do both at once. So again, Daniel, you're, you're the creative guy. How do you turn your carpet metaphor into a wave equation? Maybe there's a subwoofer in the room, or maybe somebody is waving the carpet. But what you, what you said about the, what is exchanged and what is transferred,

maybe the transfer is the classical components. I don't know which will be exchange or transfer, but one of them is the classical inscription. That's how many gigabytes were written or read to the hard drive. But what's exchanged, let's just say, are the semantics of meaning. And given an inappropriate co-preparation, maybe only one bit needs to be transferred for the exchange of a message that was already triggered to be set off. And then also this entire discussion, it kind of strikes me as like the, uh, Cartesian duality of our time, but rather than mind and body, it's the physicality of information syntax and the metaphysicality of information semantics. And that's these kind of two worlds with their own physics and metaphysics that we're trying to connect. And it's like, similarly, well, is the information, how does the information world cause something physical?

How does the physical cause something informational? And I think that the entire journey that the course and everyone is on is, uh, physics as information processing that those as a qualified and distinguished, but minimum of two set Alexi. Yeah. Thank you so much. Uh, can you hear me? Yeah. I I'm yeah.

Thinking out loud maybe, but I think the way to get closer to understanding in my head is to let go of the metaphor of asking a question, right? So if Alexi is talking to Daniel through zoom and I pose

even a closed question up or down. So I, you know, uh, formulated the question and, you know, uh, released that into zoom space and then I'm silent and I opened my listening to your answer, but you, Daniel have agency to say up or down. I did not, I do not have any effect on whether you will say up or down, right? That's the difference. So me asking you a question does not modify, uh, the answer you, you decide Alice talking to Bob, Bob has the agency to answer the question, not Alice. And that that's where I'm stuck. So I don't know if asking a question continues to be a good metaphor, prepare, preparing the instrument for measurement is, is maybe better. And, and so, but, but actually Alice is not asking Bob, Alice is acting on a qubit, right? And qubit is a, is a simple thing that, you know, can, can literally physically be up or down, I guess. And so that's, that's maybe, maybe that's the answer that Alice does not ask in Bob. Alice does something to the qubit. And because qubit can be up or down, then that's no longer asking a question, asking a question implies your, your correspondent can, can, can tell you anything.

When we're dealing with the very sophisticated process of natural language question asking, I think we're dealing, we're, we're dealing with something that that's orders, um, of composition above the kind of, the kind of binary ask. So I, I, I agree with that. Um, it's almost like, um, Alice prepares the photo receptor. So it's slate is refreshed and then it's the photon or the light emitting object that has the agency to send the photon or not. I mean, maybe the measure, the preparation occurs and then no one comes back to check whether or not agency was even exerted. And I think when we really focus on like the nano probe, like atomic force microscopy, really tracing with a super fine tip over the surface of something or scattering x-rays, when we really think about the, the, the, um, moment of information transfer, we realize like nothing comes for free. And I think that this question asking framework, other than really conveniently re-deriving quantum mechanics in one hour, incredibly helps us understand that we don't get to ask it for free in terms of time. We don't get to ask it for free in terms of energy. And we don't get to ask free floating questions where the question and or agentic responder are not involved already in a semantics. So I think it's calling attention to, the components of an experiment or a question or an opening, but agree that it's not always as simple as a binary question, but in the special, simple basal case, it does start there.

Andre, do you want to look, you want to look at any of the, um, the, uh, participant submitted questions? If there was any that struck your face.

Let me finish real quick. I have like maybe five minutes to wrap this up. Uh, so, and, and this, I think is going to at least in part answer one of Dean's concerns. So the next thing we should, uh, very briefly cover. So let's talk a little bit about, you know, holographic screens. And also, uh, this is the paper on the, you know, physical meaning of holographic principle. So let's take a look at also this picture from that paper. This is the holographic boundary and, you know, Landauer's principle and all these thermodynamic constraints on information processing necessitate, uh, the breakdown of this boundary in terms of, you know, free energy extraction sectors, memory sectors, and so on. And by the way, if you, if you read, um, the paragraphs around this picture,

there's more details on how the time QRF is induced, but we'll leave that as it is for now. Uh, Dean, what I meant by this answering your, your, your concern, your concern was that you don't see how Alice can take off her eyes from the picture. Yeah. Uh, but that's precisely what's going on here, right? There is, there is a sector that's being observed, which is a subset of these qubits that are being observed, but presumably she has the freedom to, you know, say, hey, this sector, I'm now I, I declare as my observation sector, this I declare as my memory sector, this I declare as my thermo, uh, free energy extraction sector. So it is because of these fundamental limits on the amount of information processing. It is not true that she can, she doesn't have an, an idealized, you know, you know, uh, Laplacian, if you wish, um, access to all the degrees of freedom. She can only observe finally, uh, a finite amount of stuff. So, so she, she is forced to take her eyes off at least part of the system, at least for a while. Right. Because otherwise, you know, all those energy and time constraints that Daniel was just mentioning, you know, they're not, we, you have to abide by those. So, um, so a few things that Chris mentioned in the last lecture when it came to this holographic screen is, well, number one, uh, that all the information exchange is mediated by bits, uh, here at the whole qubits at this holographic screen. This in turn enforces, um, the finiteness in the information transfer, if you wish, there is not infinitely, uh, an infinite amount of information in finite space. Even if you were to pack it at Planckian density, which is presumably the limit that black holes achieve, it's still a finite amount of information in a finite amount of. So if you have finite icons, you are taking. Uh, the other is the following. Um, and again, uh, Chris will deal with this at greater detail in perhaps lecture five, I think maybe four, but it's the notion of emergent space time, which is that A and B separately. So you think of, you know, these are presumably the internal degrees of freedom, what's going on inside of A, Alice at an abstract level, there may not be a notion of space, but, you know, the notion of space emerges at the boundary. And, uh, that's, uh, not easy to explain. I think presumably has to do with and stuff like that. Um, and I don't know the full details myself, but, but for now, that's, that's, that's one of the key things that, that, that Chris mentioned, which is that an agent, you know, may not have space, but it may emerge and the last thing, and, uh, I'm glad that to see that this was mentioned already. So it saves us time. I don't have to say it again. It'll take us a minute. This five months. And this again answers your, can you turn off your video under just so you don't have lag? Yeah. Thank you.

One second.

One second.

Yes.

Can you, yeah.

Me?

Go for it.

Okay. So this notion of, uh, Feynman diagrams, right? So what is this whole thing about unitary evolution or preparation? The, the scattering picture as Alice preparing a vector of her boundary, right? And writing it into her memory. And then

measuring it at some point in the future, right? Requires that you consider the possible what happened to that preparation. Okay. So if there's some amount of time passes, you prepared, you know, your state at time t , you look at it again, at time t plus dt , you have to take into account all the possible histories.

Clearly, I, myself of the system, or at least that part of the system, right? Dean is just saying, I prepared it here. Now it's up to B what happens until the next time I look at it, up to the environment. So I, I, I have to, in, in that unitary evolution, I consider all the possibilities, right? And that's, that's exactly, you know, the Feynman, you know, the role for. So let's just look at the conclusion.

I'll leave it there for now. No, we can open up the discussion. So there's two things that are now, where was this? Yeah. Yes. So, go ahead.

Yeah. So all I was, I understand, or I think I understand, I think I understand, that if I'm looking at on, or if I'm attending to on, I'm not looking at off. What, what I saw in that wave thing with Feynman was, I could also be looking at the switch, where the two different places of on and off exist. And when I'm looking at the entire switch, I'm not looking at one or the other. I'm looking at the, at that basal, as Daniel described, that basal opportunity of both. And, and I think that on that, I don't know how that ties in with reference frames, but on that level that says, I'm now looking at all possible. In that moment, I don't think at least from that perspective, that reference frame, I'm zooming into a on or off binary choice decision branch, I'm, I'm still at that place where the preparation allows for, for both. I'm not, I'm literally not taking my eyes off at that place. So again, I'm not arguing with you, because I what you described, and the image that you brought up, all that is, is bringing it down to a granular level and saying, if I'm paying attention to this, I can't be paying attention to that. But if I physically, from a transfer perspective, back up, so that I can see both, both possibilities at once, I don't know that I'm taking my eye off of either. That's, that's the only thing that I was bringing up. Now, of course, Chris will probably hear this and go, Dean, you got to go and rethink this whole thing through, which is great, because I'm here to learn. But that's, again, if I miss, I miss spoke about what I was trying to say, it wasn't because I, I don't see those moments when I'm directing my, my preparation over here. And that's going to give me one kind of measurement versus if I'm taking in a larger timeframe, and now or larger spatial frame or larger holographic screen, like I don't even know how to describe it, because I just don't have the words for it yet. But there you go. I just want to be clear that I'm not, I don't think I'm confused. I just want to know the difference between when I'm looking at on or off. And when I'm looking at the entire switch, that's all. I'll add one point on this slide. It goes back to some of the earliest points that Chris made about the ontological primacy of communication. Space and time arise from communicating agents. Communicating agents are not something that we have to squeeze into an a priori spatial temporal framework. So quantum mechanics, saying here on the slide, is about measurement, not mechanical motion. So we don't need to wait around or speculate as to how from the mechanical or physical hard drive. Measurement arises or happens. We start with the

measurements

occurring. And then different kinds of sophisticated agents can conceptualize of movements through spaces and times. And this also relates to Bayesian mechanics as a mechanics and a physics of movement on information geometric spaces. And so the physics can be projected by appropriately prepared

agents. But we don't necessarily need to wonder about how that agent fits within a mechanical setting.

And I guess this is where humans are with the I am a strange loop, Gertl Escherbach, gripper and gripped mystery of being able to see it multiple ways.

Andre, do you want to go to the questions?

Yeah, let me just say, are some remarks here? I think what Chris is the punching line of the paper, free energy principle for quantum systems, which is, you know, morally speaking, it's similar to the relaxation to thermodynamical equilibrium, which is that if you have an isolated bipartite quantum system a and b are going to turn u_m are going to tend towards reference frame alignment and therefore entanglement and this these two things are are basically very intimately related by aligning the reference frames you'll get a and b

will be driven towards maximal entanglement

so uh should i uh take a look at the questions and i can open this yeah

so so uh if you guys have access to the q a

uh there's been more questions in the last you know bringing you guys's attention to a few of them

uh this is where um chris links uh to the papers of robelli which i mentioned earlier in passing

when we were talking about the preparation measurement cycle and the notion of thermal time or

uh you know subjective time if you wish somebody brought up this remark that you know performance some calculations here you can relate t time to a change in u_m

uh a change in the volume of phase space or you know more precisely the the amount of available states in the state space you know with the log and whatever uh with respect to the energy right and you know you know so so this you know restates the

the idea that we're talking about earlier right like in your intrinsic uh subjective time being

this ds the e derivative a thermal time basically speaking that is uh observer dependent

so that's a a good question to uh take a look at

um there's another one what is what is meant by local free of choice uh this is a very good answer

to local here means that's on boundary right isolated uh systems are idealized by you know

definition there's always going to be an observer interacting with an environment so

a nice a purely isolated system is an idealization so what local free choice means here is exactly what what we said earlier right that you get to choose the spin axis at the boundary qubits

um

and there were some uh nice questions relating mark of blankets to the scattering picture

uh so this one for instance is also worth a look

uh

so somebody you know brought up this you know very good point about the scattering matrix
the scattering matrix is by the way you know let's not think about it mathematically just um
let's think about the picture right you you imagine two particles that are interacting
and far away they are free right they are spatially separated and interacting free particles
and somehow at some point they come to interact right and then you have to write all the finite diagrams
for all the exchange of bosons or whatever the interaction is and and that's how you would compute
the amplitudes for the scattering process right how those two particles that were you know separated came
to interact and then they go off to infinity again uh chris here brings up the point that in the sense
the mark of blanket is you know in the opposite direction right you see a marker blanket in at least in the
graph theory sense you start with a node and the marker blanket is just the set of nodes that
separate in terms of conditional probability that node from everyone else right and then you can have edges
which are going to be directly uh causal networks so acyclic
no no no cycles directed edges that's that's how the graph looks
and you you build up so essentially you're taking the opposite um picture right you start with a node
which in this case would be the interaction and then you build up to infinity as opposed to in the
scattering picture which is like you start with particles not interacting at infinity you see how they go off to
infinity again
and then the interaction is in the middle uh so that obviously the answer is is a bit more uh precise but it's
worth taking a look at
and this is a an interesting one too of a symmetrical input input what bandwidth so again if you think of it as a
graph nothing is constraining you to say that you're going to have
more incoming errors than outcome errors into the boundary right that that's just the way it is you know if you
think of it a graph there is no symmetry imposed the only natural sense of symmetry would be energy
conservation and that's exactly what you know chris says here right which is that um
and i'm not sure how you related to a conservation of not quite information but you know i wouldn't know how
to phrase it myself
i would just take a look at the question but basically no there's a priori nothing that necessitates symmetry in
input output in markov blankets
but if you are to talk about any sort of symmetry the natural language would not be in terms of incoming
versus
zaka mineros but rather energy conservation which we we saw is is you know a close casting of information
transfer right because of
the minimal requirements for action to rewrite a qubit
and the last question that i would like to bring your you guys's attention to i mean all of them are great
uh but we have finite resources and finite times on internet uh would be
would be uh this one okay and i'm not gonna go into too much detail but this is again about
the local wait no one second
um
so it turned on my camera by the way dino go for it

and also people people can see all the questions on the website right and submit their own as well
yes for sure uh so and now i lost track of the question that i wanted to show you guys
i think it's this one over here yes uh yes exactly this one so by the way all of them are good even
this one the holographic principle does it make it easier to compute any physics problems well please take
a look at chris's answer but all i can say is that these ideas of holography are i'm not an expert i'm
not a condensed matter physicist but they're they're used all over physics nowadays not necessarily just in
high energy
i know that condensed matter theorists use them all the time so the answer to this is
uh an emphatic yes the holographic at least adacft is used all over but again i'm not a physicist myself
um so this is the last question i wanted to bring your attention to and uh here chris uh this this may
be the one worth taking a look at in terms of uh in order to answer all uh you know the earlier
speculations that we had about time internal clock here somebody asks about um
synchrony and
you know just the perception of time with respect to external clocks i suppose
and chris here answers that at the end of the day any even if you're using an external clock
everything that you perceive outside at the end of the day has to be in terms of an internal clock
any information incoming you're gonna have to you know make causal sense of it you're gonna have to
write it into a memory in terms of an internal clock so so you're never going to be able to
ease so to speak increase your precision of the passage of time by taking an external clock
even if you think it's more precise than anything you have inside i think that's basically what his
answer amounts to and this is morally related to you know all this frame halting problem stuff where because
i'm never as a system you're never going to give uh an infinitely precise you know i'm speaking very loosely
here an infinitely precise description of yourself you're you're never gonna morally speaking again
that's that's what it amounts to you're never going to be able to outdo your internal clock by
by trying to leverage something external like like you know an atomic clock that even if you think it's more
precise at the end of the day your own subjective perception of information will depend on internal clocks
so we can uh open up things any concluding remarks and any other alexi or dean or anyone else who may
have joined i don't have anything else to say at this point i mean where where or when are we we had
session one
with a real big historical overview section two we focused on the measurability question which is answered
with
the question and we talked about some of the constraints of question asking namely that it takes time it takes
energy you have to ask it to somebody or something on an interface and it has to have a semantic quantum
reference frame and this is dovetailing with a free energy principle of course because the free energy
principle
as it says in the syllables to what does the free energy principle apply and chris says everything
measurable that's why it's a theory of every thing that we choose to measure model not ontological
statements about what is and then where are we heading into in session three

have you looked ahead under or do you want me to read what's in the syllabus

uh please do yeah yeah i think basically uh my short answer to that and possibly my full answer also is that it's just going to take care of the loose ends in this lecture two so this lecture two is a warm-up to the to the real thing which we need to talk more seriously about reference frames to in order to get the precise uh picture of this quantum measurement process yeah yeah it says that session three will introduce the idea of quantum reference frames and the representation using hierarchies of binary classifiers these formal structures provide a semantics for measurement and hence provide the basis for a theory of meaning for interacting agents the language of qrfs allows a particularly straightforward and intuitive definition of variational free energy and so allows a fully general quantum formulation of the fep we will see that the fep is a classical limit on the principle of unitarity the fundamental principle of quantum theory uh by the way uh daniel can i i'm not sure if you're busy but i just got an email from chris i think he may be watching us and i think he's just confused about the link so chris i don't know if you're there still but um do you think it's worth if we send him the link real quick if you can come for a few minutes yeah or i will oh i'll send chris the um the zoom link now so yeah hold on hold on to your qubit chris we're coming yeah

so let's let's let's give him a few minutes and maybe he's still watching and he'll he'll answer the questions better than i could well while daniel is sending him sending him a a lifeline can i just from the from the the rookie perspective on this session one seemed like there was a big time window like the historical piece over a using an external clock and a timeline this one felt like the parabola and the upper down and the entanglement was almost instantaneous in terms of that plank length that we were really holding up as comparative to that timeline a chronological timeline and so i'm guessing that now that we've established those two as different where does that go but again that's just the rookies view of things hey chris welcome yeah hi this is a very interesting discussion um and it's uh given me some pointers as to what to emphasize at the beginning of session three uh but i i can say in response to dean's question that the time when alice isn't looking is the time during which bob is making his measurement and then his next preparation step

so if you if you consider again that slide that shows uh sort of the fineman representation of what's happening as this unitary uh flow that's what bob is doing and i think andrew said that oh from alice's point of view bob is just some quantum system and she's banged on that system by asking a question

and now that system is doing something or other that she can't see until it it i.e bob bangs on his side of the screen

and she gets some information by making a measurement uh so she she may be not looking for a very short period of time

uh right i mean according to her internal clock she's not looking for just the time it takes for one of her internal ticks to take place uh but uh in in general uh we're not saying anything about how long that time is right if you think of this in a scattering theory way uh then we're talking about the time between two protons leaving the rings on the uh large hadron collider to the time that all the stuff that comes out of their

collision is measured by the detectors which is a nanosecond or something like that some very short period of time

uh so just think of that interval as the the measure of the intrinsic

uh time dependence external time dependence of the interaction someone has to do something and then someone else has to do something and there's a gap uh between what the first person does and what they see

in response oh let's see what's the other question uh the the other central question that came up today was the question of why observing something requires action and of course in the in the classical picture observing something doesn't require any action at all uh the the event just happens and this is the the fundamental if you will distinction between classical and quantum theory is that making an observation in in in quantum theory is taking an action on the world and there's a lovely uh metaphor that's that's not correct in detail but it's it's very useful i think conceptually uh which heisenberg came up with to try to explain his uncertainty principle to his classical colleagues he said look suppose i'm trying to to to to see a very small particle and measure its motion oh i have to have a microscope to see this particle and to measure its motion i need to to light it up i need to shine some light on it and those photons of light are actually impacting the thing i'm trying to see

so the the the act of observation itself requires banging something into an object

and when we're just looking out at the world it's the sun that's banging the light into the object

not us but we can we can sort of merge those two systems together because it's the sun that's taking the action that lets us get some information but the sun is perturbing the things that we're looking at

by banging photons into them um and i think that that picture of of heisenberg's is what uh he used to to actually derive the uncertainty principle in terms of momentum and uh position because banging something into something is transferring momentum to it photons transfers of momentum to the tiny particle in the microscope and that perturbs its position on the microscope stage which is precisely what heisenberg is trying to measure so that's his his point in that analogy observation itself always takes action and that's why observation and preparation are duals in quantum theory because they each require a minimum

one unit of action and i think andrew you said the the duality can be thought of in terms of time and variance and that's exactly correct preparation is just measurement run backwards right and vice versa and if you let me say a few words i think this also answers my own confusion and the one that alexi the question that alexi posed earlier which is this duality between preparation and measurement in your picture that you're bringing up chris with heisenberg by shining photons on a particle that i'm trying to observe um these pictures of preparation and measurement are dual because in in doing so i'm constraining the system in trying to measure it i'm constraining it so it is so in me trying to gather information by shining photons on it what i'm really doing is i'm constraining it i'm just saying you can only take this this you know momenta or or position is would that be correct uh yeah in fact you're in the microscope example it's particularly clear you're you're bouncing a photon off of the object and then it's that photon that you're detecting right oh there was one other

question that uh can be answered quickly it was the question about uh what's the reference frame that uh alice is using when she's uh measuring up or down measuring her spin up or down and of course there's there's several reference frames involved here oh so let's work inwards uh in a real system alice is sitting here she's got a stern gerlock apparatus which is a vertical magnet apparatus and she's measuring the spin of an electron that's that's traveling through this stern gerlock device and what the stern gerlock device does is is orient the spin uh with a magnetic field that's that's not homogeneous so it ends up pointing physically up uh as it comes through the apparatus so how is that apparatus oriented i mean what defines the up direction for that apparatus well in the practical setting what defines that apparatus orientation is the earth's gravitational field right we're sitting we're in we're here in the laboratory and the earth's gravitational field tells us what direction is down it's toward the center of the earth and so up is away from the center of the earth and that's what defines the axis for the apparatus which is a great big piece of metal so it's very responsive to the gravitational field so now we can ask about alice right alice is also embedded in this gravitational field and she also can detect the gravitational field she can tell which way is down toward the center of the earth by the the force acting on her feet and that measurement is taking place in alice's brain so in alice's brain there's a representation of which way is up and that's the internal reference frame of alice that lets her do the experiment you know if if she was freely floating in space but the apparatus was anchored to the gravitational field she wouldn't be able to tell that it was anchored to the gravitational field that's einstein's the point that einstein makes in all of those experiments where he has observers flying around in spacecraft uh that they have to have their own reference frame uh what what einstein didn't emphasize was that reference frame is in their head it's encoded by their nervous system uh so uh i i think it was daniel who said this exactly correctly that uh the physical object uh in her head the the actual neural network that tells her which way the gravitational field is uh is itself a semantic object i mean that's that's what gives the word up and down meaning for alice and what is what is what do i mean when i say the word meaning i mean it lets her take some action um semantics boils down to actionability if it gives things significance and this this was the point that that bateson was making when he said differences that make a difference uh for an organism uh something is meaningful if it actually makes a difference for what i can do uh in this world that i'm trying to survive in oh so i'll i'll leave it at that but this is this is very very useful for uh leading off the next session so thank you very much for for all that discussion awesome thank you i guess if i could say one thing here is that uh let me scroll back up to the three points early in the slides differences that make a difference would tell you that you require action to to perturb the system and measure and then the semantics or the contextuality that may be encoded in the in the in the nervous system would tell you what is the difference that is being made would that be correct yeah right what what tells you that there's a difference there it's it's it's what what you're able to detect and represent internally oh it's almost like step one do differently step two you have to ask it to a different thing and then three you need the semantic frame that makes a and b different

difference differences all the way down um well we will transcribe this and fix the few patchy moments
enter we'll work together on updating the transcript so that everybody who wants to be well prepared
for the measurement of the third lecture will be absolutely in place
in place thank you chris for joining that's a fun fun cap to this discussion so all right thank you see
you all next time bye
thank you

Thank you.